

Technology Stack

Date	15 March 2026
Team ID	EDUGENIE-01
Project Name	EduGenie – AI-Powered Educational Assistant
Maximum Marks	2 Marks
Team Lead	Pavan Varma Uddaraju
Members	Voruganti Krishna Sanket Naresh C Humaira Tasneem Syed Sai Subash Raja Kosaraju

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5 + CSS3 (Vanilla)	Lightweight and fast-loading UI with no dependency on heavy frameworks. Ensures cross-browser compatibility and runs efficiently on resource-constrained devices including Mac M1. Responsive design for desktop and mobile.
2	Backend / Server-Side	Python 3.10+ with FastAPI	FastAPI provides high-performance async REST API support with automatic Swagger documentation. Python is ideal for AI/ML integration and the team has existing expertise. Supports all modules: /qa, /explain, /quiz, /summarize, /learn/recommendations.
3	Database / Data Storage	SQLite (Dev) / PostgreSQL (Prod)	SQLite requires zero configuration for local development and testing. PostgreSQL is used for production due to its robustness, scalability, and support for complex queries. Both are compatible with Python's SQLAlchemy ORM.
4	Cloud / Hosting / Deployment	Local / Docker / GitHub Codespaces	Optimized to run locally on Mac M1 and standard PCs. Docker containerization ensures consistent deployment across environments. GitHub Codespaces enables cloud-based development and testing without infrastructure setup.
5	Version Control & CI/CD	Git + GitHub	GitHub enables seamless team collaboration, pull request reviews, and version history. GitHub Actions can be configured for automated testing and deployment pipelines (CI/CD). All team members use VS Code or PyCharm with Git integration.

6	Third-Party APIs / Other Tools	Google Gemini API (gemini-pro)	Powers all AI capabilities — intelligent Q&A;, concept explanation, quiz generation, text summarization, and personalized learning path recommendations. Chosen for its high-quality generative responses, free-tier availability, and Python SDK support.
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